Maria Luísa Rocha

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EDUCATION

Brown University
Ph.D. Candidate in Earth, Environmental, and Planetary Sciences
Providence, RI | Expected Graduation May 2027

Master of Science in Environment and Society and Earth, Environmental, and Planetary Sciences, 3.8/4.0 GPA
Providence, RI | Graduated May 2024

UFRJ - Federal University of Rio de Janeiro
Bachelor of Science in Meteorology, 3.75/4.0 GPA
Rio de Janeiro, Brazil | Graduated July 2021
Honors: *cum laude*

CEFET/RJ - Federal Center for Technological Education of Rio de Janeiro
Technical Degree in Meteorology
Rio de Janeiro, Brazil | Graduated December 2016

RESEARCH INSTITUTIONS AND INDUSTRY

Brown University, Graduate Research Student
Providence, RI | September 2022 – Present

- Conduct research on polar lows, Arctic climate dynamics, machine-learning applications in Arctic sea ice, and climate-related maritime hazards.
- Design and conduct regional atmospheric modeling experiments using the Weather Research and Forecasting model, including WRF and Polar WRF.
- Develop and analyze numerical-model ensembles using reanalysis, observational, and climate datasets.
- Use high-performance computing systems to run regional atmospheric simulations and process large meteorological datasets.
- Work with high-performance computing environments at Brown University and NCAR.
- Collaborate with interdisciplinary teams on climate data analysis and modeling.

Norwegian Meteorological Institute (MET Norway), Research Visitor
Oslo, Norway | October 2026 – December 2026

- Study how machine-learning weather forecast models work, their limitations, and how they are being used and advanced in research for the Nordic Seas.
- Learn from operational forecasters how weather risks are assessed and communicated for the Nordic Seas.

National Center for Atmospheric Research, Summer Research Visitor
Boulder, CO | Summers 2023 – 2026

- Conducted regional atmospheric-modeling research focused on simulations and sensitivity experiments for Arctic weather systems.
- Used NCAR high-performance computing resources to run and analyze Polar WRF simulations and ensemble experiments.
- Collaborated with NCAR scientists on atmospheric-model configuration, experimental design, and interpretation of model results.
- Presented ongoing research at the NCAR Happy Hour Seminar on June 3, 2026.

Climatempo and State Environmental Institute of Rio de Janeiro, Meteorologist
Rio de Janeiro, Brazil | January 2022 – September 2022

- Delivered daily weather forecasts and analyses for national and regional audiences.
- Analyzed weather patterns using meteorological models to predict severe weather events that could endanger the population.
- Communicated meteorological findings to non-technical audiences through reports and briefings on the “Flood Warning” State website and Social Media.
- Monitored C-band radars in two distinct locations in the state of Rio de Janeiro, ensuring their operational functionality through remote maintenance. Developed custom products based on radar outputs to support meteorological analyses and decision-making.

CPTEC/INPE, Research Fellow
Cachoeira Paulista, Brazil | February 2018 – August 2021

- Performed statistical evaluation of precipitation forecasts and simulations from the ETA model for seasonal and subseasonal timescales.
- Focused on the São Francisco River Basin, Brazil, to improve predictive accuracy and understand hydrological patterns.
- Contributed to national research projects aimed at enhancing the reliability of atmospheric and hydrological models.

Nimbus Meteorologia, Intern and Meteorologist
Rio de Janeiro, Brazil | November 2020 – November 2021

- Developed Python scripts for data mining and technology development initiatives within the company.
- Analyzed meteorological models and presented results as visualizations and reports.
- Conducted a three-year study on lightning trends across Brazil using data from the Geostationary Lightning Mapper (GLM) sensor on GOES-16.
- Provided daily weather forecasts for regional projects across various Brazilian states.

CONFERENCES AND PRESENTATIONS

Rocha, M. L. (2026, June 3). Polar Lows and their implication for Arctic Maritime Safety [Seminar presentation]. *NCAR Happy Hour Seminar, National Center for Atmospheric Research,* Boulder, CO.

Rocha, M. L., & Lynch, A. (2025, May). Assessing the Impact of Polar Lows on Ship Icing Risks in the Arctic [Talk presentation]. *American Meteorological Society's 18th Conference on Polar Meteorology and Oceanography at the 2025 AMS Denver Summit*, Denver, CO.

Perkins, E., Lynch, A., Bailey, D., Solvyeva, V. & **Rocha, M. L.** (2025, May). Predicting Snow Structures Relevant to Reindeer Husbandry [Talk presentation]. *American Meteorological Society's 18th Conference on Polar Meteorology and Oceanography at the 2025 AMS Denver Summit,* Denver, CO.

Rocha, M. L., Lynch, A., & Bergen, K. (2025, April). Enhancing Sea Ice Concentration resolution in a Northern Sea Route strait using a Generative Adversarial Network [Poster presentation]. *DEEPS Dive 2025*, Brown University, Providence, RI.

Rocha, M. L., Lynch, A., & Bergen, K. (2023, December). Enhancing CMIP6 model's sea ice concentration with machine learning on the Northern Sea Route [Poster presentation]. *American Geophysical Union 2023*, San Francisco, CA.

Rocha, M. L., Chan, C. S., Dereczynski, C. P., & Resende, N. C. (2020, November). Evaluation of subseasonal to seasonal forecasts of the Eta-BESM model in the São Francisco River basin in El-Niño years [Talk presentation]. III Symposium on the São Francisco River Basin, Online, Brazil.

Rocha, M. L., Chan, C. S., & Satyamurty, P. (2020, August). Evaluation of subseasonal forecasts of the Eta model in detection of the start of the rainy season [Poster presentation]. National Institute for Space Research Scientific Initiation Seminar, São José dos Campos, SP, Brazil.

Rocha, M. L., Resende, N. C., Chan, C. S., Dereczynski, C. P., & Satyamurty, P. (2019, November). Evaluation of subseasonal to seasonal precipitation simulations of the Eta model for the São Francisco River basin [Talk presentation]. XXIII Brazilian Symposium on Water Resources, Foz do Iguaçu, PR, Brazil.

Rocha, M. L., Resende, N. C., Chan, C. S., Dereczynski, C. P., & Satyamurty, P. (2019, October). Detection of the start of the rainy season in the São Francisco River basin from the forecasts of the Eta model nested to the BESM model [Poster presentation]. XVIII Meeting of Postgraduate Students in Meteorology from CPTEC/INPE, Cachoeira Paulista, SP, Brazil.

Pristo, M. V., Souza, P. R., Carvalho, N. O., Dereczynski, C. P., & **Rocha, M. L.** (2018, October). Study of frontal systems associated with heavy rain events in the city of Rio de Janeiro [Poster presentation]. XIII Argentinian Meteorology Congress, Santa Sé, Rosário, Argentina.

PUBLICATIONS

Rocha, M. L. & Lynch, A. (in prep). *Quantifying Marine Cold-Air Outbreak Influence on polar low development in the Barents Sea.*

Lynch, A. & **Rocha, M. L.** (In Review). *Controls on Arctic Maritime Cyclone Intensification: Insights from an Updated Climatology*.

Rocha, M. L., & Lynch, A. (2026). Influence of sea ice and Greenland topography on polar lows cases in the Greenland Sea. Monthly Weather Review. https://doi.org/10.1175/MWR-D-26-0049.1

Rocha, M. L., Lynch, A., & Bergen, K. (2025). *Enhancing sea ice concentration resolution in a Northern Sea Route strait using a generative adversarial network.* Journal of Geophysical Research - Machine Learning and Computation 2, no. 1. https://doi.org/10.1029/2024JH000281

Rocha, M. L., Resende, N. C., Dereczynski, C., Vasconcellos, F. C., & Chou, S. C. (2023). *Evaluation of precipitation simulations at the subseasonal range in the São Francisco River Basin, Brazil*. Anuário do Instituto de Geociências, 46, 1–12. https://doi.org/10.11137/1982-3908_2023_46_59421

Chan, C. S., Resende, N. C., **Rocha, M. L.**, Dereczynski, C. P., Gomes, J. L., & Sueiro, G. (2018, September). From subseasonal to seasonal forecasts over South America using the Eta Model [Conference paper]. Numerical Weather and Climate Modeling: Beginnings, Now, and Vision of the Future, Belgrade, Serbia.

BOOK CHAPTERS

Bailey, D. A., Li, X., Lynch, A. H., Kao, B., King, T., Perkins, E. T. D., & **Rocha, M. L.** (In press). Chapter 3. The Changing Arctic Cryosphere. In A. H. Lynch, L. C. Smith, & L. W. Brigham (Eds.), The Many Arctics in 2050: Converging Pressures and Diverging Futures. Cambridge University Press.

Şalap-Ayça, S., Chen, N., Aryal, S. C., **Rocha, M. L.**, Bennett, M., & Tingstad, A. (In Press). *Chapter 9. Making the Arctic Legible. In A. H. Lynch, L. C. Smith, & L. W. Brigham (Eds.),* The Many Arctics in 2050: Converging Pressures and Diverging Futures. Cambridge University Press.

Silva, L. M. da, **Rocha, M. L.,** Machado, B. L. dos S., Foligno, P. P., Carvalho, L. A. S., & Ferreira, R. D. (2023). *Meteorology and mountaineering: Dialogue between meteorology students and practitioners of the sport (in Portuguese).* In R. Guedes (Ed.), Brazilian Science: Multiple Perspectives – Nutrition, Biology, Biomedicine, and Environment. Even3. https://doi.org/10.29327/cb--nutricao-biologia-biomedicina-meio-ambiente-1.599744

UNDERGRADUATE RESEARCH MENTORSHIP

Perlis, J., **Rocha, M. L.**, Lynch, A., Şalap-Ayça, S. (2026). *Spatial and Temporal Trends in Arctic Weather Hazard*. Summer Research Symposium.

Perlis, J., **Rocha, M. L.**, & Lynch, A. (2025). *Polar Cyclone Trends and Maritime Risks Near Greenland*. Summer Research Symposium, Brown Digital Repository. https://doi.org/10.26300/g7d4-v149

Mogridge, E., **Rocha, M. L.**, & Lynch, A. (2025). *Evaluating Climate Model Skill in Simulating Arctic Sea Ice Response to Volcanic Forcing*. Summer Research Symposium, Brown Digital Repository. https://doi.org/10.26300/f9dd-dg42

TEACHING

Graduate Teaching Assistant, *Introduction to Atmospheric Dynamics*
Instructor: Amanda Lynch, Spring 2024 and Spring 2026

- Supported course delivery by assisting students with atmospheric science concepts.
- Facilitated discussions on dynamic processes in Earth's atmosphere and clarified complex topics during office hours.

Undergraduate Teaching Assistant, *Statistics Methods Applied to Atmospheric Sciences*
Instructor: Renata Libonati, Spring 2020

- Clarified class topics during office hours and grading.

Undergraduate Teaching Assistant, *Introduction to Atmospheric Sciences*
Instructor: Claudine Dereczynski, Spring 2018, and Leonardo Aragão, Fall 2018

- Clarified class topics during office hours.
- Organized a weekend field trip to Itatiaia National Park, Brazil, with meteorological instruments.

FELLOWSHIPS

Jarislowsky Graduate Fellowship (2026): Brown University, USA.

WORKSHOPS AND TRAINING

The Sheridan Center for Teaching and Learning (2025): Brown University, USA.

Scalable and Computationally Reproducible Approaches to Arctic Research Course (2025): Arctic Data Center (ADC), USA.
WRF Tutorial Winter Session (2023): National Center for Atmospheric Research (NCAR), USA.
VI Worketa Short Course (2019): Brazilian Center for Weather Forecasting and Climate Studies (CPTEC), Brazil.

PROFESSIONAL ENGAGEMENT

World Meteorological Organization Research Board Meeting
Geneva, Switzerland | October 2024
Attended the Eleventh Session of the WMO Research Board (RB-11), engaging with discussions on international priorities and coordination in weather, climate, and Earth-system research.

Arctic Circle Assembly 2026
Reykjavik, Iceland | October 2026
Attended the Arctic Circle Assembly annual meeting, engaging with discussions on Arctic priorities to engage all stakeholders in a changing Arctic, including state representatives, scientists, policy makers, indigenous populations, citizens, and layers.

SKILLS

Technical Skills
Proficient in programming languages and data analysis: Python, Fortran 90, ArcGIS
Experienced with climate modeling tools: WRF, Polar WRF, Eta Model
High-performance computing: Slurm, PBS, Linux-based computing environments, parallel model simulations, and large meteorological datasets
Climate and meteorological data analysis: reanalysis, observational, ensemble, and numerical-model datasets
Machine learning for environmental applications: GANs, SR-GAN, statistical forecasting

Language Skills
Bilingual in English and Portuguese

Soft Skills
Strong analytical and problem-solving abilities
Effective team collaboration in research projects

COMMUNITY AND DEPARTMENTAL SERVICE

Graduate Student International Mentor
Department of Earth, Environmental, and Planetary Sciences, Brown University | 2024-2025 Academic Year
- Organized events to foster interaction and create a sense of community among international graduate students within the department.
- Provide the international DEEPS community with valuable information about life as an international graduate student.

Graduate Student Social Chair
Department of Earth, Environmental, and Planetary Sciences, Brown University | 2023-2024 Academic Year
- Organized events to foster interaction and create a sense of community among graduate students within the department.
- Collaborated with faculty and students to plan and execute social and professional development activities.

Graduate Student Assistant
International Students Orientation, Brown Graduate School | May 2025 – September 2025
- Coordinated events to welcome and integrate new international students into the Brown University community.
- Invited graduate students from various programs to deliver short instructional talks to incoming students.
- Led and supervised campus tour groups, emphasizing best practices and ensuring a smooth introduction to Brown University.
- Designed and organized the campus tour program, enhancing the onboarding experience for new students.